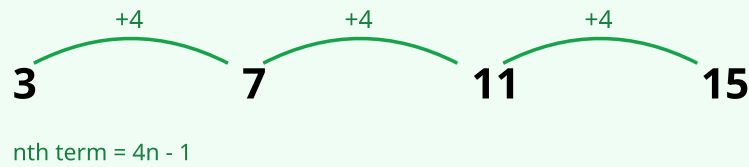


Arithmetic Sequences — nth Term

AQA · KS3 Year 7–8

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE



Arithmetic sequence: a constant difference is added each step.

COMMON MISTAKES

Writing $d + n$ instead of dn .

Wrong sign on the constant.

Confusing term number with term value.

Not checking that n is a positive whole number.

METHOD

- 1 Find the common difference d
- 2 Rule starts with dn
- 3 Constant = first term - d
- 4 $\text{nth term} = dn + \text{constant}$
- 5 Check with $n = 1, 2, 3$

WORKED EXAMPLE

Find the nth term of 3, 7, 11, 15, ...

$d = 4$ $4n$; constant = $3 - 4 = -1$.

$\text{nth term} = 4n - 1$ (check $n = 1$ gives 3)

VISUAL AID

The arcs between terms show the constant difference (+4 each step). The rule $4n - 1$ gives the value of the term in position n .

1. (2 marks)

Write the next two terms: a) 2, 5, 8, 11, ... b) 20, 17, 14, ...

2. (2 marks)

Find d and the nth term: a) 4, 6, 8, 10, ... b) 1, 5, 9, 13, ...

3. (3 marks)

Find the nth term: a) 5, 8, 11, 14, ... b) 7, 12, 17, 22, ... c) 10, 8, 6, 4, ...

4. (3 marks)

$\text{nth term} = 4n - 1$. a) 10th term? b) 20th term? c) Which term is 51?

5. (3 marks)

Is the number in the sequence? a) 100 in $3n + 1$? b) 50 in $4n - 2$? c) 30 in $5n + 2$?

6. (3 marks)

Theatre: row 1 = 15 seats, row 2 = 18, row 3 = 21. a) Rule for row n . b) Seats in row 10.

7. (4 marks)

$\text{nth term} = 6n - 4$. a) First three terms. b) 50th term. c) Show 201 is not a term.

8. (5 marks)

A: $4n+1$, B: $3n+7$. a) First five terms of each. b) Common value? c) Solve $4n+1 = 3n+7$ and explain.